

Replication Plan

Joint Emulation of Earth System Model Temperature–Precipitation Realizations with
Internal Variability: fldgen v2.0

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Auto-generated Replication Blueprint

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Paper Overview

This paper describes `fldgen` v2.0, an R package for generating spatially and temporally correlated realizations of temperature and precipitation fields that emulate the output of Earth System Models (ESMs). The method decomposes ESM output into a forced response (fitted via global-mean temperature scaling patterns) and residual variability, then generates new variability realizations preserving space–time correlation structure and cross-variable (temperature–precipitation) correlations using empirical orthogonal functions (EOFs) and a vector autoregressive (VAR) model.

Detailed Workflow

Phase 1: Package Installation and Data Acquisition

1.1 Install `fldgen` v2.0 from GitHub.

1.2 Download ESM training data.

- CMIP5/CMIP6 model output (temperature and precipitation fields).
- Global-mean temperature trajectories from simple climate models (e.g., Hector, MAGICC).

1.3 Prepare input data in the required NetCDF format.

Phase 2: Pattern Scaling (Forced Response)

2.1 Fit scaling patterns.

- Regress gridded temperature and precipitation fields against global-mean temperature.
- Obtain per-grid-cell scaling coefficients (pattern fields).

2.2 Compute residuals.

- Subtract the scaled forced response from the ESM output to obtain variability residuals.

Phase 3: Variability Model (EOF + VAR)

3.1 Joint EOF decomposition.

- Concatenate temperature and precipitation residual fields.
- Compute EOFs (PCA) to reduce dimensionality; retain leading K modes.

3.2 Fit VAR model to the EOF time series to capture temporal autocorrelation.

3.3 Generate new realizations.

- Draw innovations from the fitted VAR noise distribution.
- Propagate through the VAR model to produce new EOF coefficient time series.
- Project back to physical space via EOF patterns.
- Add forced response to obtain full temperature/precipitation fields.

Phase 4: Validation

4.1 Compare emulated statistics to ESM output.

- Spatial correlation structure (variograms).
- Temporal autocorrelation functions.
- Cross-variable (T–P) correlations.
- Marginal distributions at selected grid cells.

4.2 Reproduce paper figures.

Datasets

Dataset / Source	Type	Location / Notes
fldgen test/example data	Bundled	Included in the fldgen package https://github.com/jgcri/fldgen
CMIP5/CMIP6 ESM output	Public	https://esgf-node.llnl.gov/search/cmip6/
Hector global-mean temperature	Public	https://github.com/JGCRI/hector
MAGICC pathways	Public	https://magicc.org/

Models, Tools, and Software

Programming Languages

- **R 4.0+** — fldgen is an R package.

Libraries and Frameworks

Tool / Library	Version	Purpose / URL
fldgen	v2.0	Core field generation package https://github.com/jgcri/fldgen
ncdf4 (R)	latest	NetCDF file I/O
vars (R)	latest	Vector autoregressive model fitting
raster / terra (R)	latest	Geospatial data handling
ggplot2 (R)	latest	Visualization
Hector	latest	Simple climate model for global-mean temperature https://github.com/JGCRI/hector

Hardware Requirements

- Standard workstation; EOF computations on global grids benefit from ≥ 16 GB RAM.